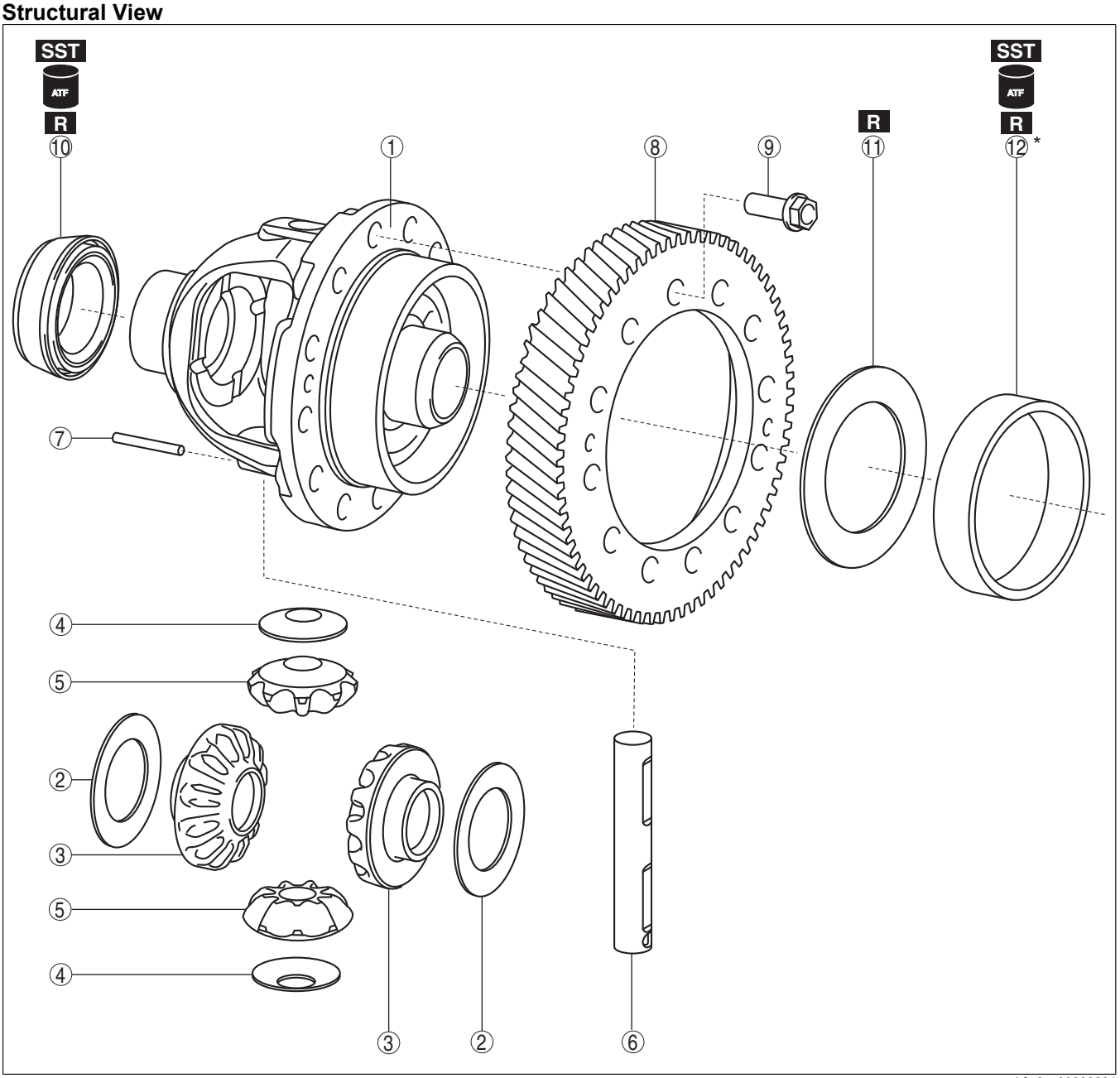
RING GEAR AND DIFFERENTIAL ASSEMBLY [EW6A-EL/EW6AX-EL]

id051700664114

Structural View



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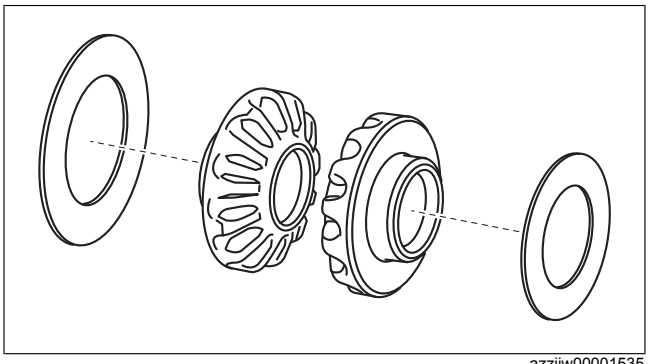
1	Differential gear case
2	Thrust washer (selection)
3	Side gear
4	Thrust washer
5	Pinion gear
6	Pinion shaft
7	Pin
8	Ring gear

9	12 bolts (M11×1.0 bolt, length to approx. 30 mm {1.2 in})
10	Taper roller bearing (inner diameter approx. 46 mm {1.8 in})
11	Shim
12	Bearing race*

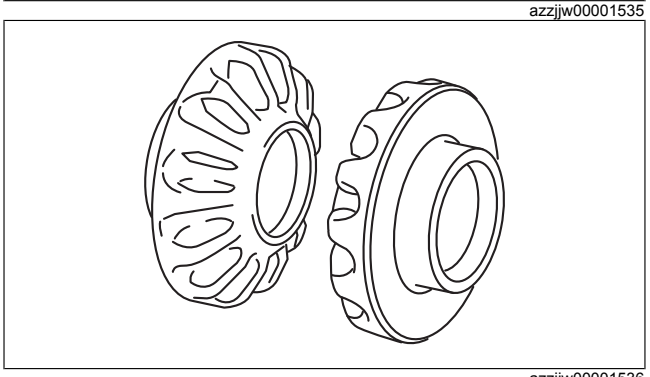
* : When replacing the tapered roller bearing on the transaxle side, replace the bearing race with a new one.

Assembly Procedure

1. Assemble the thrust washers to the side gears.

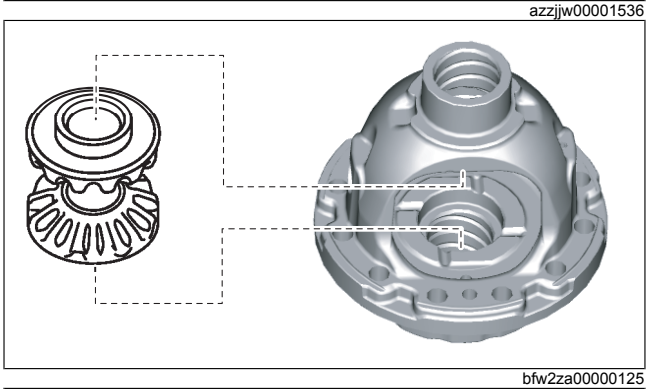


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azzjjw00001536

2. Assemble the side gears which have the thrust washers assembled to them.

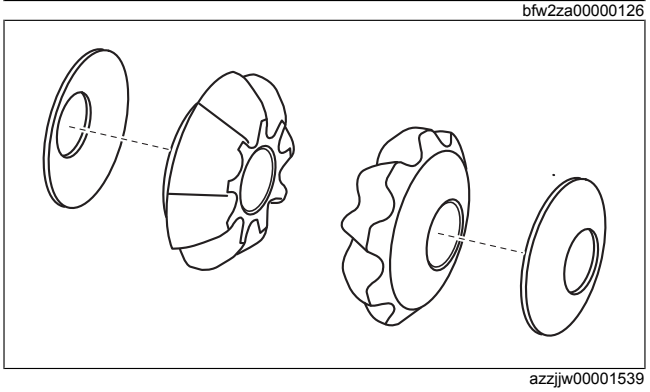


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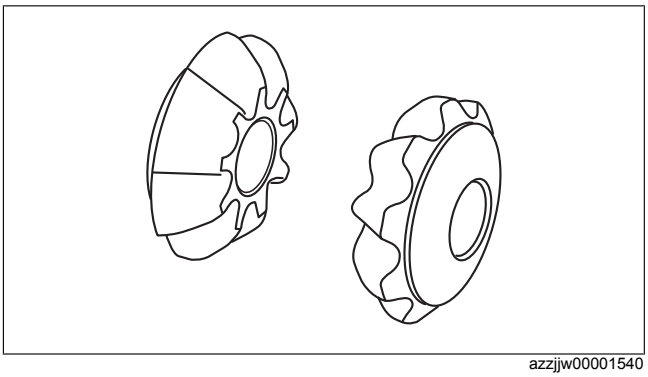


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3. Assemble the thrust washers to the pinion gears.

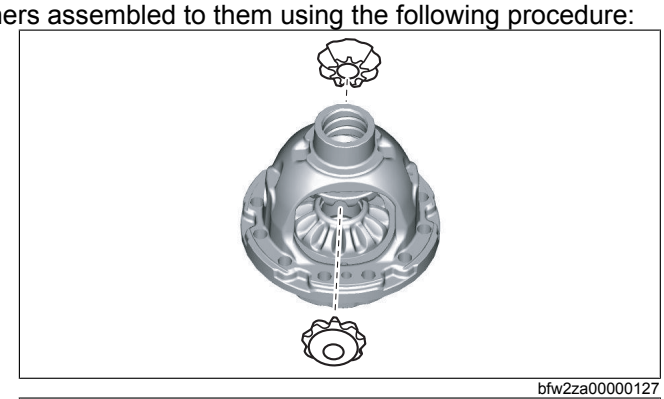


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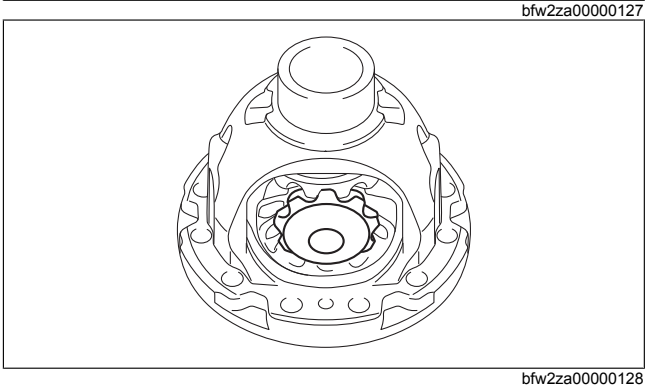


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4. Assemble the pinion gears which have the thrust washers assembled to them using the following procedure:
   (1) Assemble the pinion gears which have the thrust washers assembled to them.



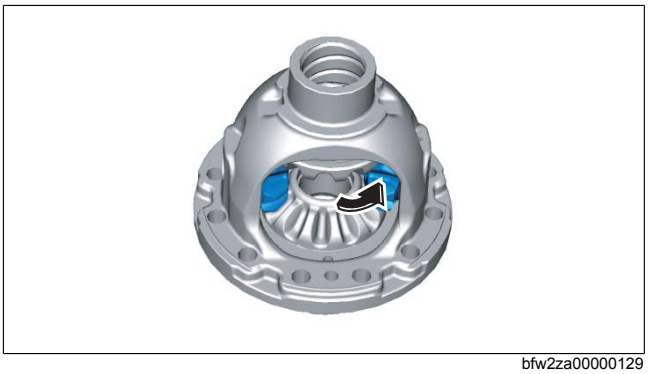
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bfw2za00000128

   (2) Rotate the pinion gears so that the pinion shaft holes of the differential gear case and the pinion gears are aligned as shown in the figure.

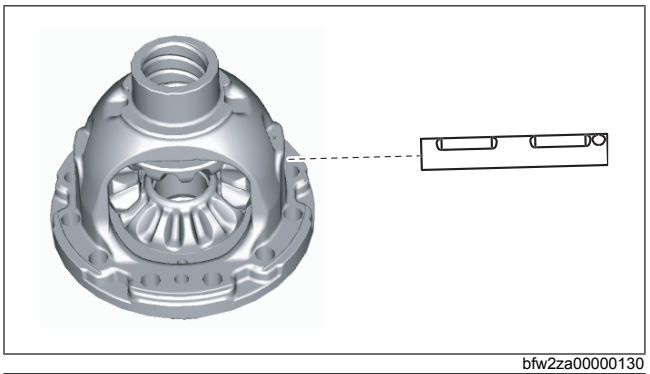
Note
- If the pinion shaft holes of the differential gear case and the pinion gears are not aligned, remove the pinion gears, change the side gears combination, and reassemble.



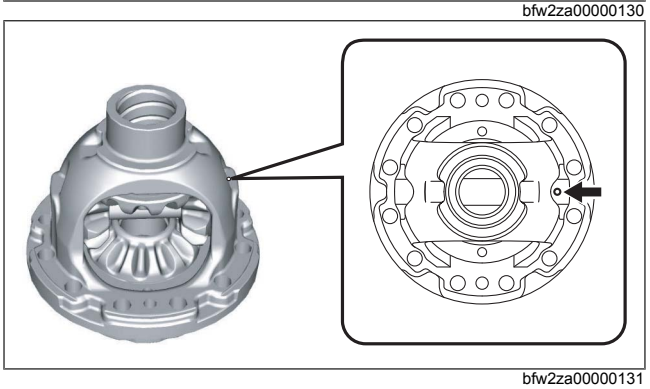
bfw2za00000129

5. Assemble the pinion shaft.

Caution
- **Assemble the pinion shaft so that the pin holes of the differential gear case and the pinion shaft are aligned.**



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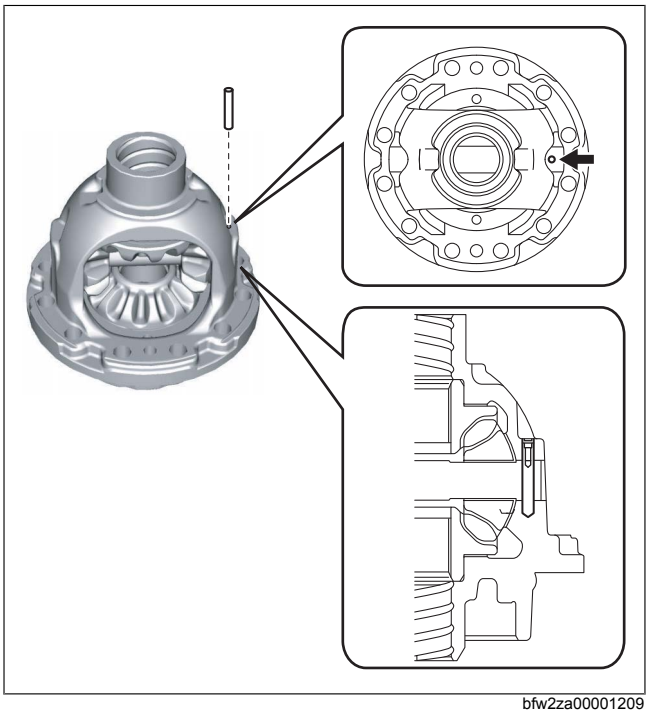
bfw2za00000131

6. Assemble the pin to the position shown in the figure using a pin punch.

Note
- Use a pin punch with an end outer diameter of 3.5 mm {0.138 in} or more, and within 5 mm {0.197 in}.

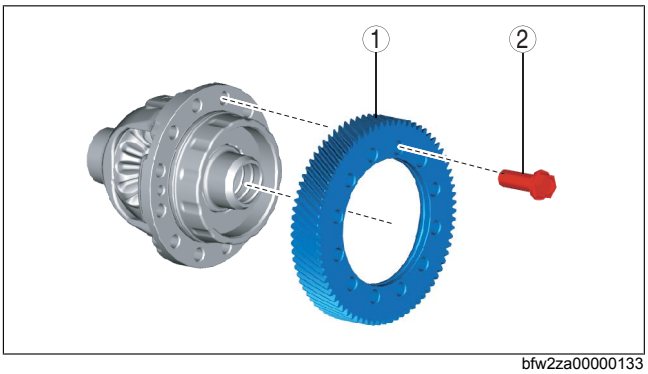
7. After assembling the pin, crimp the pin hole ends (2 locations) of the differential gear case.
8. Install the screw used when disassembling to the pin, and verify that the pin cannot be pulled out when pulling the screw.

Note
- size: M3 x 0.5, length: 10 mm {0.394 in} or more



bfw2za00001209

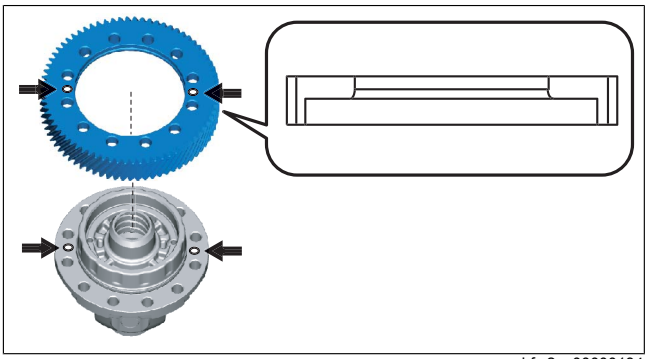
9. Assemble the ring gear using the following procedure:

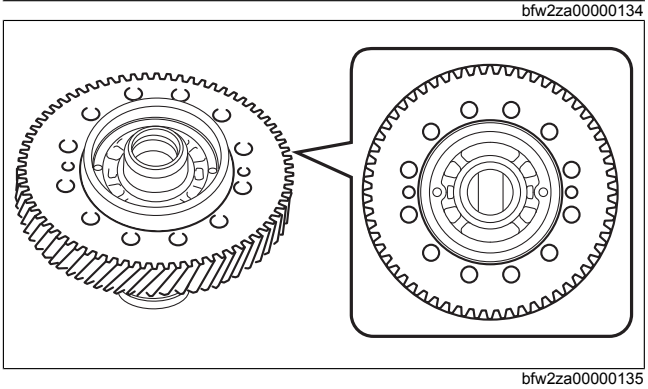


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(1) Assemble the ring gear.

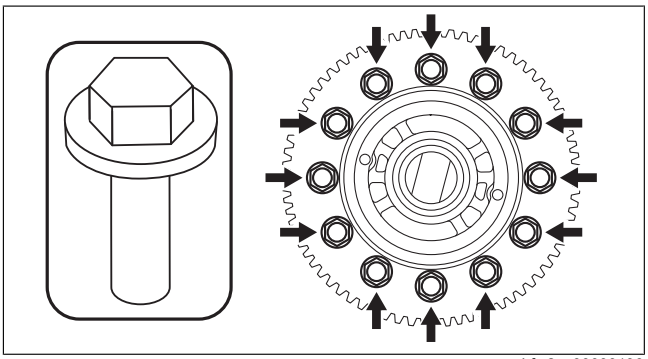
Note
- Assemble the ring gear so that the differential holes and ring gear holes shown in the figure are aligned.


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bfw2za00000135

(2) Assemble and temporarily tighten the bolts to the positions shown in the figure.

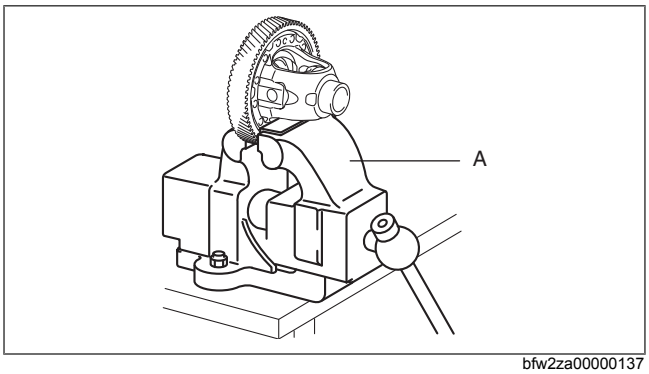
Note
- Bolt size: M11×1.0 bolt, length to approx. 30 mm {1.2 in}


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(3) Secure the ring gear and differential in a vise.

Caution
- **Insert a protective plate between the vise and the part so as not to damage the part.**

A : Vise

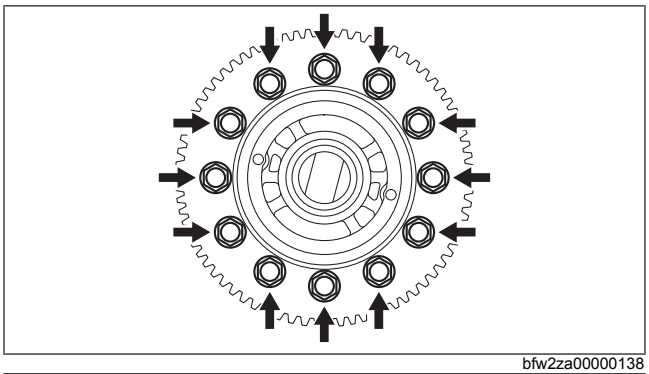


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(4) Tighten the bolts shown in the figure.

Note
- Change the vise securing position of the ring gear and differential and tighten all of the bolts shown in the figure.

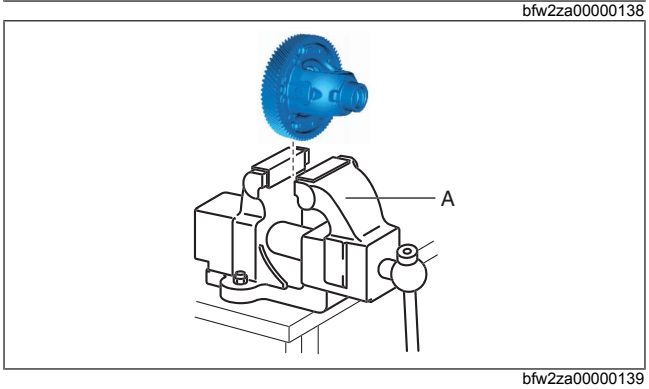
Tightening torque
90.2—98.0 N·m {9.20—9.99 kgf·m, 66.6—72.3 ft·lbf}


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(5) Remove the ring gear and differential from the vise.

A : Vise

1	Ring gear
2	12 bolts (M11×1.0 bolt, length to approx. 30 mm {1.2 in})

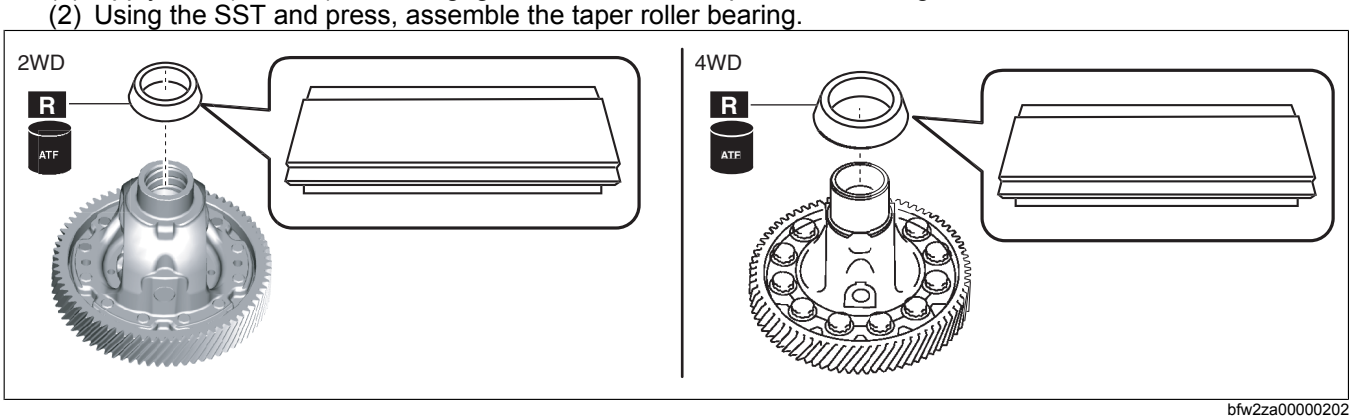


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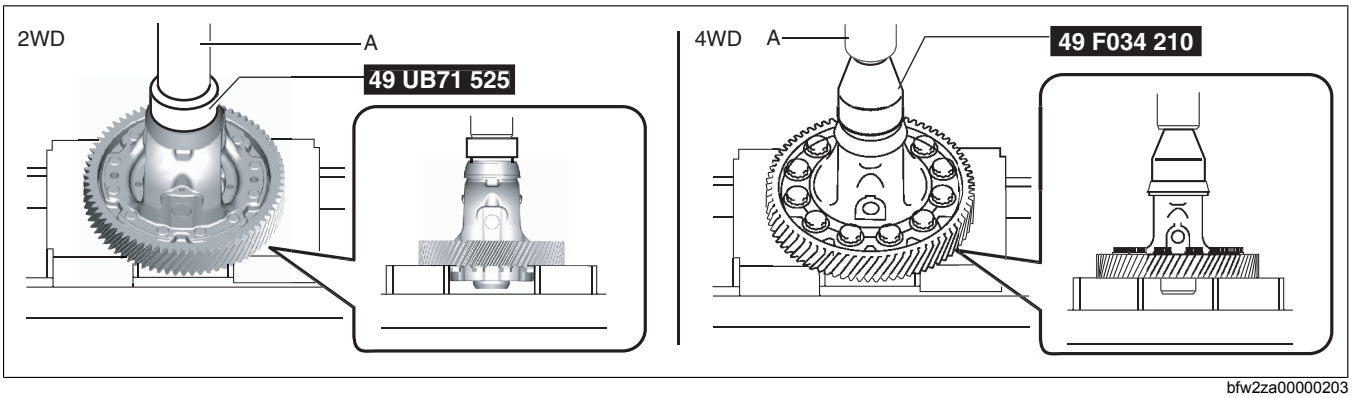
10. Assemble the taper roller bearing (converter housing side) using the following procedure:

Note
- Bearing race size: Inner diameter approx. 45 mm {1.8 in} (2WD)
- Bearing race size: Inner diameter approx. 55 mm {2.2 in} (4WD)

(1) Apply ATF (ATF FZ) to the engagement area of the taper roller bearing and differential.
(2) Using the SST and press, assemble the taper roller bearing.



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bfw2za00000203

A : Press

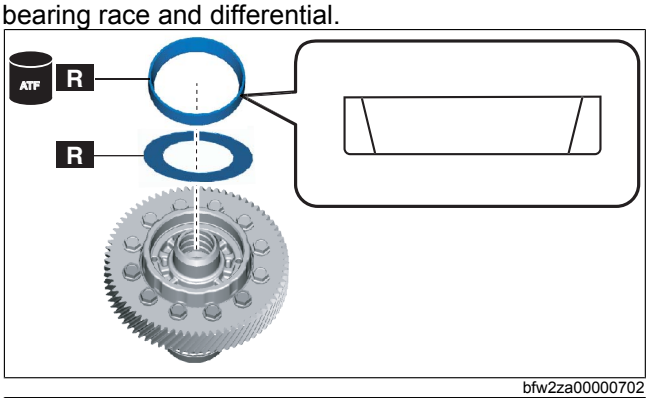
11. Assemble the shim, bearing race using the following procedure:

Note

- Bearing race size: outer diameter approx. 90 mm {3.5 in}

(1) Apply ATF (ATF FZ) to the engagement area of the bearing race and differential.

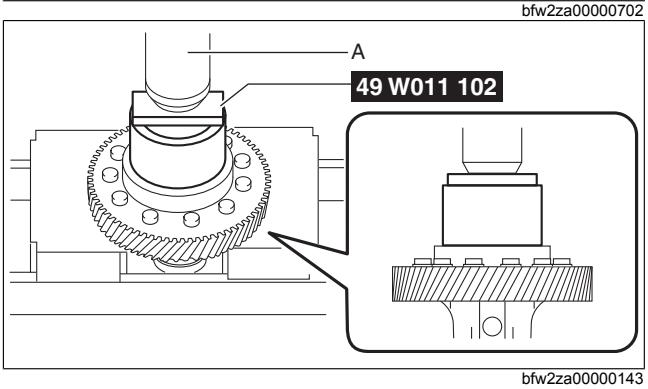
(2) Using the SST and press, assemble the bearing race.



bfw2za00000702

A : Press

12. Perform the differential backlash measurement/adjustment. (See DIFFERENTIAL BACKLASH MEASUREMENT/ADJUSTMENT [EW6A-EL/EW6AX-EL].)



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